

Seminar Presentation



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Previous Studies

Tuning code smell prediction models: A replication study (ICPC 2024)

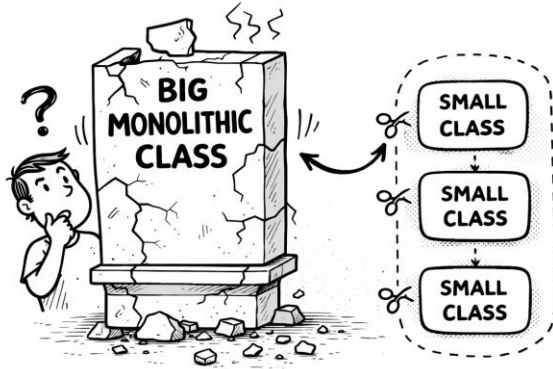
Evaluating the effectiveness of LLMs in fixing maintainability issues in real-world projects (SANER, 2025)

Marv: A manually validated refactoring dataset (FORGE 2025)

Current Goals and Motivation

- Few studies explore the use of LLMs for architecture smells tasks.
 - “*Exploring Architectural Smells Detection Through LLMs*” (ECSA’25)
 - “An exploratory study on architectural smell refactoring using Large Language Models” (ICSA’25)
- Architectural smell detection using LLMs is still underexplored.
- We’re exploring the LLM-based refactoring in architectural level.
 - Architecture Smell Detection Study
 - Refactoring Architecture Smells

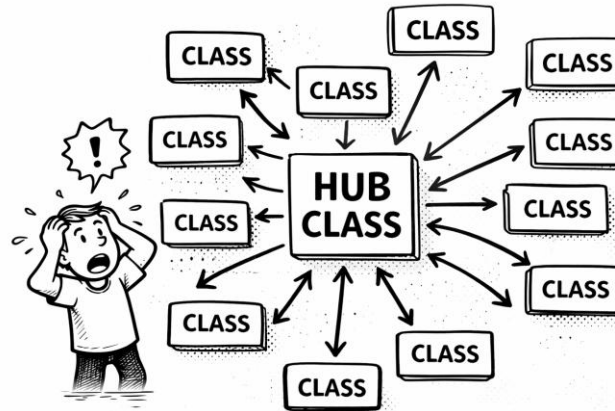
Architecture Smells



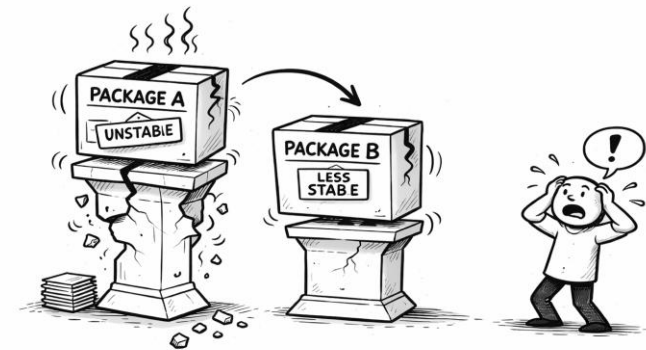
Insufficient Modularization



God Component



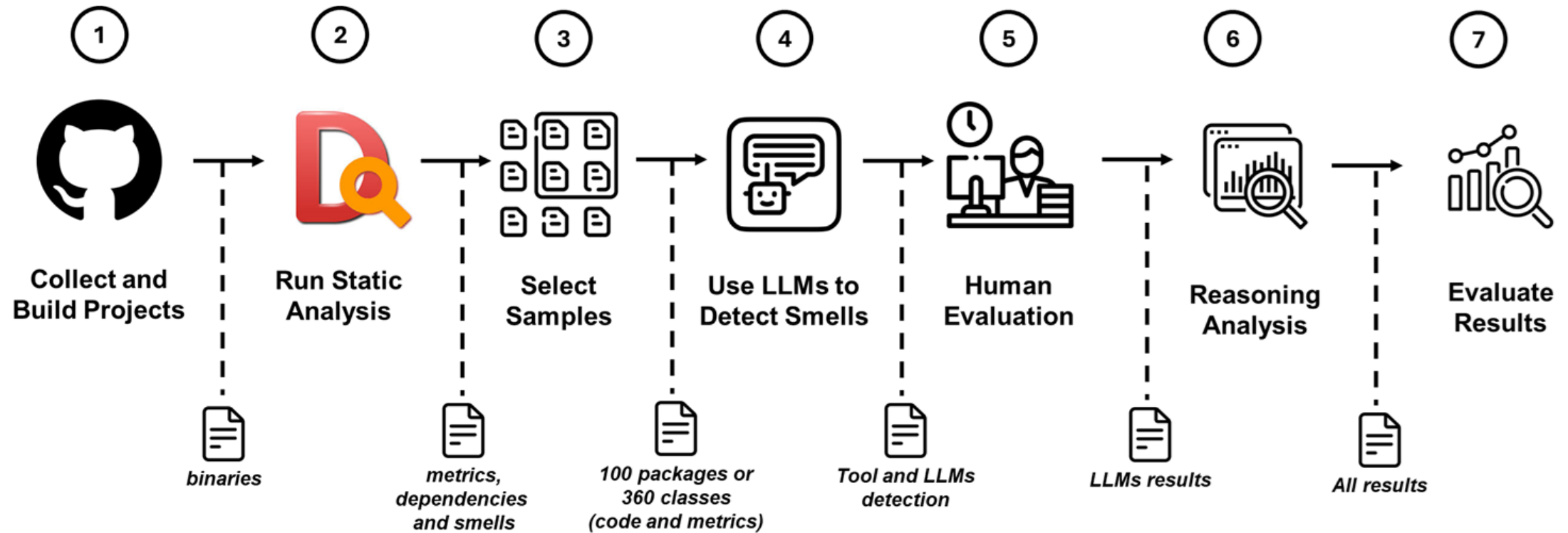
Hub-like Modularization



Unstable Dependency

Detection Study

Detection Study: Design



Detection Study: Models and Prompting

- GPT 5 mini
 - context size ~400k
- DeepSeek V 3.2
 - context size ~100k
- Qwen3 Coder 420B
 - context size ~100k

Task

You are a software engineering expert in software refactoring.

Your task is to detect the smell:

[smell name].

Definition: *[smell literature definition]*.

Constraints

[constraints list]

Input

{input data}

Output

```
{  
  class/package: "name",  
  detection: true/false,  
  justification: "Reasons for the detection result"  
}
```

Detection Study: Architecture Representation

Raw Code

```
var perc = 99.0, wmin = 1920, hmin = 1080, w, h, w1, h1, ratio;
var FromDoc = open ( File ("D:\FromMacro.psd"));
var IntoDoc = open ( File ("D:\IntoMacro.psd"));

app.preferences.rulerUnits = Units.PIXELS;
w = FromDoc.width.value;
h = FromDoc.height.value;
ratio = h/w;
app.activeDocument = FromDoc;
activeDocument.activeLayer = activeDocument.layers[0];

var shapeRef =
[ [ Math.floor ((w-1920)/2), Math.floor ((h-1080)/2) ],
  [ Math.floor ((w-1920)/2)+1920, Math.floor ((h-1080)/2) ],
  [ Math.floor ((w-1920)/2)+1920, Math.floor ((h-1080)/2)+1080 ],
  [ Math.floor ((w-1920)/2), Math.floor ((h-1080)/2)+1080 ] ];

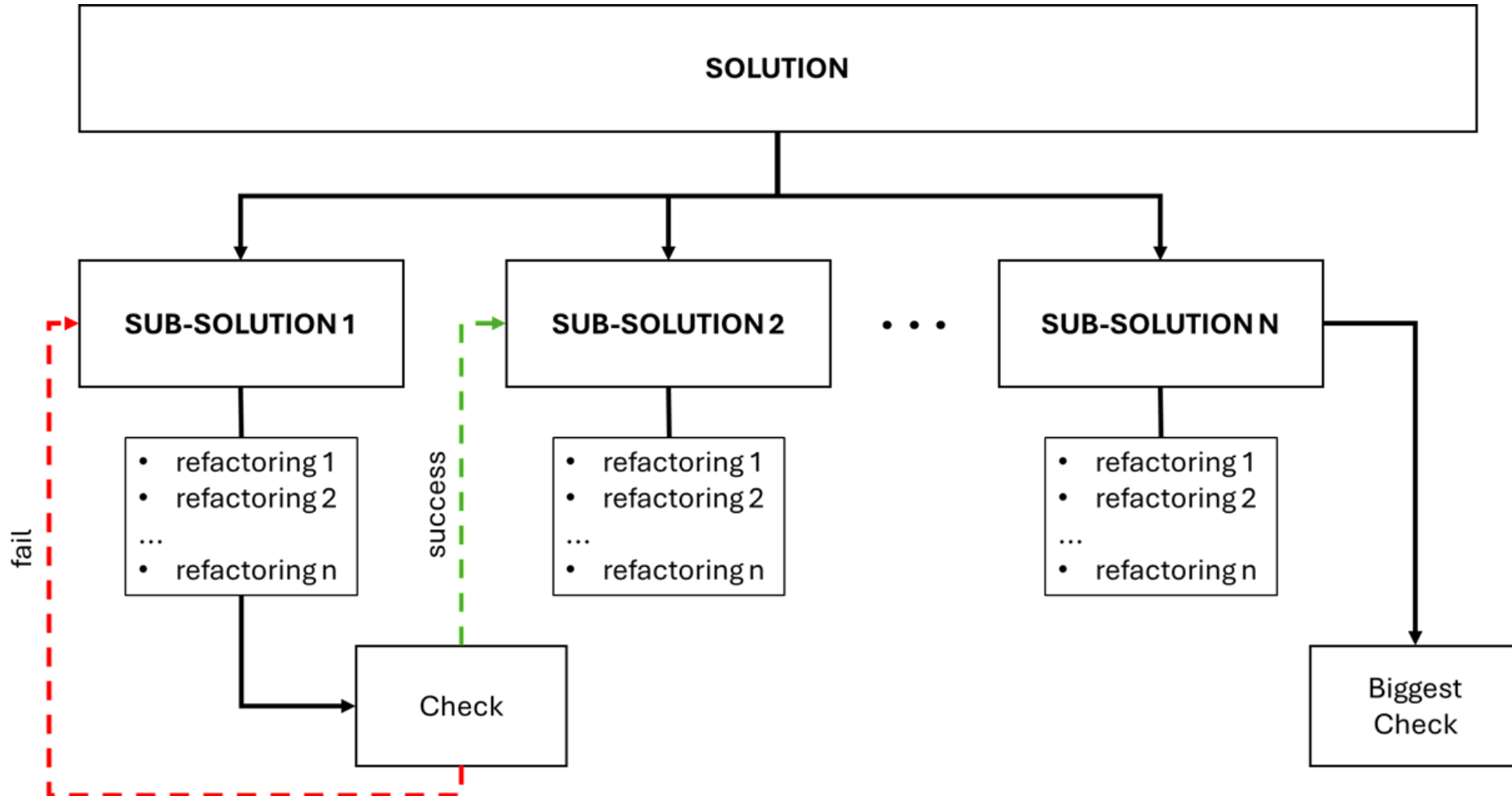
app.activeDocument.selection.select ( shapeRef, SelectionType.REPLACE );
app.activeDocument.selection.copy ();
app.activeDocument = IntoDoc;
activeDocument.activeLayer = activeDocument.layers[0];
IntoDoc.paste ();
```

Metrics & Dependencies

```
{
  "project": "jsoup",
  "summary": {
    "total_packages": 8,
    "total_classes": 200
  },
  "packages": [
    {
      "package": "org.jsoup.examples",
      "metrics": {
        "num_classes": 4,
        "loc": 190,
        "Ce": 4,
        "Ca": 0
      },
      "classes": [
        {
          "package": "org.jsoup.examples",
          "class": "ListLinks",
          "file": "data/repositories/jsoup/src/main/java/org/jsoup/ex",
          "metrics": {
            "nof": 0,
            "nopf": 0,
            "nom": 3,
            "nopm": 1,
            "loc": 34,
            "wmc": 8,
            "nc": 0,
            "ncc": 0
          }
        }
      ]
    }
  ]
}
```

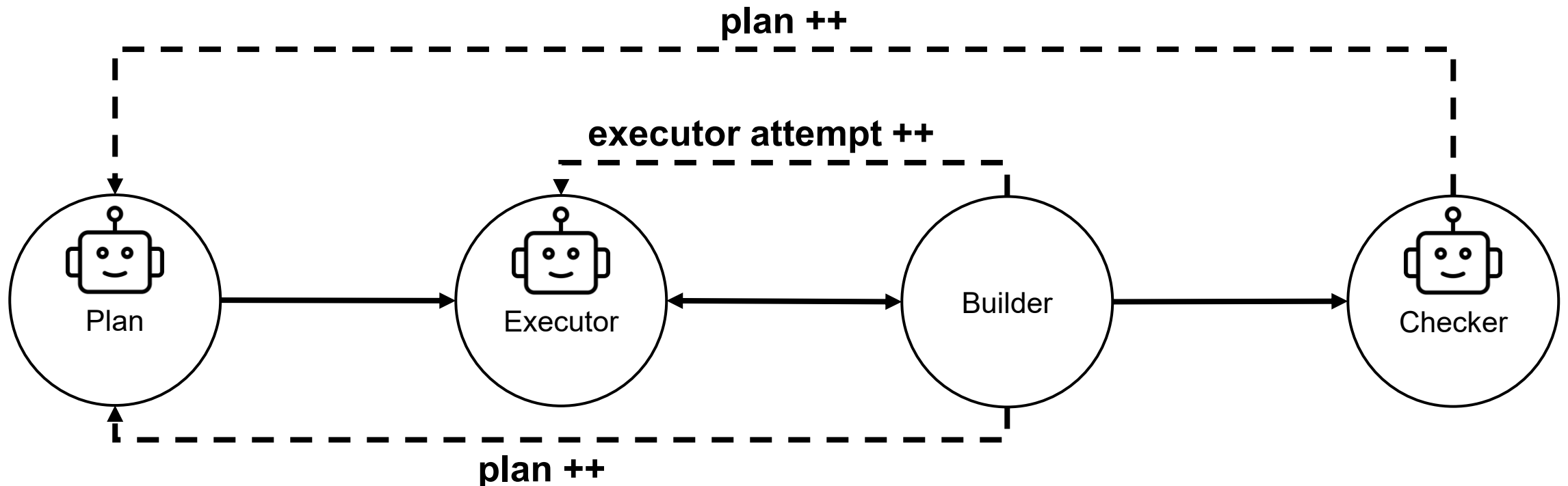

Refactoring Study

Refactoring Study: Approach



Refactoring Study: Framework

- Three tries per sub-solution.
- If sub-solution fails 3 times, replan.
- At maximum 5 plans.



Current Progress

- Detection Study
 - Quantitative Analysis: comparison between LLMs and (1) Designite Detection and (2) Human Evaluation
 - Qualitative Analysis: architectural elements used by LLMs to justify their detection
- Refactoring Study
 - Successful refactoring cases for Insufficient Modularization and Hub-like Modularization.
 - God Component in progress.

Thank you!

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